Status	Finished
Started	Thursday, 22 May 2025, 4:55 PM
Completed	Thursday, 22 May 2025, 5:10 PM
Duration	15 mins 31 secs
Marks	12.00/12.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Complete

Mark 1.00 out of 1.00

Which kind of transformation is *guaranteed* to leave the set of optimal policies unchanged?

- [ ] a. Clip to 1 all the rewards
- [ ] b. Any affine transformation of immediate rewards only.
- [x] c. A strictly monotonic transformation of the total return.
- [ ] d. Squaring the reward at each step

Question 2

Complete

Mark 1.00 out of 1.00

The slides note that *potential-based reward shaping* is mathematically equivalent to...

- [ ] a. doubling the learning rate
- [ ] b. value-function normalization
- [ ] c. reward clipping
- [x] d. initializing the Q-values with the potential function

Question 3

Complete

Mark 1.00 out of 1.00

Compared with a feed-forward network (FNN), an LSTM used for contribution analysis tends to produce *lower-variance* differences in consecutive predictions Direct Credit Assignment.
What statistical property of the LSTM's prediction errors explains this?

- [x] a. Errors are *highly correlated*, so they cancel in the difference
- [ ] b. Errors have heavier tails, making large differences rare
- [ ] c. Errors are i.i.d. across time
- [ ] d. Errors are biased toward earlier time-steps

Question 4

Complete

Mark 1.00 out of 1.00

The slides note that potential-based shaping is *equivalent to Q-value initialisation* $Q^{'}(s,a) = Q(s,a) + \Phi(s)$Q'(s,a)=Q(s,a)+Φ(s) Direct Credit Assignment.
What practical risk does this equivalence highlight when Φ is *learned online* with TD-errors derived from sparse, delayed rewards?

- [x] a. Bias is re-introduced because Φ itself must be bootstrapped from delayed feedback
- [ ] b. The learning rate must be halved to keep Q-updates unbiased
- [ ] c. Shaping may over-count rewards, double-updating the same transition
- [ ] d. Eligibility traces converge to zero faster, causing under-estimation

Question 5

Complete

Mark 1.00 out of 1.00

According to R. Sutton and Minsky, what is the problem of credit-assignment?

Select one or more:

- [ ] a. Assigning credit or blame for overall episodes that contributed to the expectation of the return.
- [ ] b. Identifying models that get high rewards
- [x] c. Assigning credit or blame for overall outcomes to each of the decisions that contributed to those outcomes
- [ ] d. Identifying actions that led to high rewards

Question 6

Complete

Mark 1.00 out of 1.00

Which of the following sentences are true

Select one or more:

- [x] a. Variance in MC increases with the delay of the reward
- [ ] b. TD methods corrects the bias faster than MC
- [x] c. TD(λ) assigns credit faster than TD(0)
- [ ] d. Variance in TD(0) increases with the delay of the reward

Question 7

Complete

Mark 1.00 out of 1.00

If we make the expected future rewards equals to zero (i.e. via reward redistribution), the Q-values are just the average of the immediate reward. However, why this is not possible in an MDP setting?

Select one or more:

- [ ] a. Because in order to have future expected reward to zero, states have to be enriched with the past, and therefore, breaks the Markov assumption
- [ ] b. It is completely possible in an MDP setting.
- [x] c. Because in this case (future expected reward equal to zero), the reward depends also on the previous state and action, and therefore, is not Markov.
- [ ] d. Because in this case (future expected reward equal to zero), the reward is obtained before we actually get in in the environment, and therefore, is not causal and breaks the Markov property.

Question 8

Complete

Mark 1.00 out of 1.00

What is the definition of a sequence-Markov decision process (SDP), as seen in class?

Select one or more:

- [ ] a. A sequence-Markov decision process (SDP) is defined as a decision process which is equipped with a **Markov reward** and has **Markov policy** but a **transition probabilities that is not required to be Markov**
- [ ] b. A sequence-Markov decision process (SDP) is defined as a decision process which is equipped with a **Markov policy** but a **transition probabilities and reward that is not required to be Markov**
- [x] c. A sequence-Markov decision process (SDP) is defined as a decision process which is equipped with a **Markov policy** and has **Markov transition probabilities** but a **reward that is not required to be Markov**
- [ ] d. A sequence-Markov decision process (SDP) is defined as a decision process which is equipped with a **Markov reward** and has **Markov transition probabilities** but a **policy that is not required to be Markov**

Question **9**

Complete

Mark 1.00 out of 1.00

What is the dfinition of **return-equivalent** SDPs?

Select one or more:

- [ ] a. Two SDPs are return equivalent if they differ only in their reward distributions but have the **same return per episode**
- [x] b. Two SDPs are return equivalent if they differ only in their reward distributions but have the **same expected return**
- [ ] c. Two SDPs are return equivalent if they differ only in their reward distributions but have the **same expected return per episode**

Question **10**

Complete

Mark 1.00 out of 1.00

What is the **explaining away problem** in MDPs, when a function g is used to predict the return based on the sequence of actions and states?

Select one or more:

- [x] a. Since later states are used for return prediction, earlier actions causing reward are missed.
- [ ] b. Previous actions can explain away future states, breaking the Markov property
- [ ] c. Since the sequence of states and actions are mutually independent, the prediction needs every component to explain the return.
- [ ] d. In an MDP, one only need the difference of states actions to make a prediction and not the whole sequence.

Question **11**

Complete

Mark 1.00 out of 1.00

What of the following statements are true when refering to differences between LSTM and Feed Forward networks (FFN) for making future predictions?

Select one or more:

- [x] a. LSTM will only learn to store information if a state-action pair has a strong evidence for a change in the expected return.
- [x] b. FFNs are not suited for processing sequences.
- [x] c. LSTM tends to have smaller prediction errors since reuses past information
- [x] d. FFNs are prone to prediction errors since they have to extract information from every state to make a prediction in every timestep
- [x] e. Prediction errors of LSTMs are much more likely to cancel via prediction differences. Since consecutive predictions of LSTMs rely on the same internal states, they usually have highly correlated errors.

Question 12

Complete

Mark 1.00 out of 1.00

What is the main difference between sensitivity analisys (SA) and contribution analysis (CA)?

Select one or more:

- [ ] a. **SA** is gradient-based, while **CA** is not
- [ ] b. **CA** focuses on how a small change in the input change the output, and **SA** focuses on how the input is responsible of the output
- [ ] c. **CA** is gradient-based, while **SA** is not
- [x] d. **SA** focuses on how a small change in the input change the output, and **CA** focuses on how the input is responsible of the output